Energy-Oriented Representation of General Relativity

Mathematical Working Basis for the Hypothesis "Universe = E"

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Status: Conceptual hypothesis; no alternative theory of gravitation

Part I: Basic Concepts and Fundamental Field Equations

1. Objective and Methodological Delimitation

   General Relativity (GR) describes gravitation through the geometry of spacetime. The distribution of energy, momentum, pressure, and mechanical stresses is summarized in the energy-momentum tensor (T_mu_nu). The present work investigates the extent to which the established equations of GR can be systematically presented in an energy-oriented form. The foundational hypothesis is: Universe = E Here, matter is interpreted as a specific bound and organized energy state. The mathematical reformulation of established equations is initially not intended to introduce new physical laws. Three levels are distinguished:

2. Established physics: experimentally verified interrelationships and mathematical equations of GR.

3. Energy-oriented representation: mathematically equivalent notations, for example via E_0 = M * c^2.

4. Own interpretation: the speculative assumption that energy distribution could represent a more fundamental physical description than the geometric interpretation of spacetime. An equivalent notation in itself is not proof of the third level.

5. Conventions and Basic Quantities Unless otherwise noted, the signature (- + + +) and standard SI units are used for the following equations.

Symbol	Meaning	Unit
E	Energy	Joules (J)
E_0	Rest energy of a body or energy equivalent of a mass parameter	Joules (J)
M	Mass or suitable gravitational mass parameter	Kilograms (kg)
rho	Mass density	kg/m³
epsilon	Energy density	J/m³
p	Pressure	Pascals (Pa), equal to J/m³

G	Newtonian constant of gravitation	$m^3/(kg \cdot s^2)$
c	Speed of light in vacuum	m/s
g_mu_nu	Metric tensor of spacetime	coordinate-dependent
T_mu_nu	Energy-momentum tensor	dependent on index and coordinate convention

Greek indices like mu, nu run over the four spacetime dimensions. A repeated upper and lower index implies summation in standard tensor notation.

Equation 1 – Mass-Energy Equivalence Established form: E_0 = M * c^2 Energy-oriented usage: M = E_0 / c^2 Meaning: The rest mass of a body corresponds to a rest energy. For a suitable total energy or mass parameter, this relation can be used to express the quantity in energy units. Status: Established physics; algebraic reformulation. Limit: Not every mass appearing in GR is readily the rest mass of an individual body. For composite systems, gravitational binding energies, and dynamic spacetimes, it must be clarified which energy or mass parameter is meant. Relation to hypothesis: The equation provides an established basis to express mass parameters through energy equivalents. It does not prove that matter consists exclusively of photons or that energy is ontologically superior to spacetime.

Equation 2 – Conversion of Mass Density into Energy Density Established form: epsilon = rho * c^2 Inverse representation: rho = epsilon / c^2 Meaning: Energy density corresponds to mass density multiplied by c^2, provided rho describes the associated rest mass per volume here. Status: Established relationship for this definition; direct transformation. Limit: In relativistic systems, pressure, flow, radiation, and other energy contributions can be important. The complete gravitational source is therefore not always describable by a single scalar mass density. Relation to hypothesis: Energy density epsilon can be used as a preferred quantity without changing the mathematical content of the underlying relationship.

Equation 3 – Energy Content of a Volume Formula: E = Integral over V of (epsilon(x) dV) For homogeneous energy density: E = epsilon * V Meaning: Total energy results from integrating energy density over a volume. Status: Established definition under specified conditions. Limit: In curved spacetime, volume element, reference frame, and the definition of energy must be carefully established. Local integration of an energy density cannot automatically be equated with a globally conserved total energy of the entire spacetime. Relation to hypothesis: The equation illustrates how a local energy distribution can be described mathematically. It does not yet introduce a new mechanism of gravitation.

3. The Energy-Momentum Tensor The energy-momentum tensor T_mu_nu is a central quantity of GR. It describes not merely energy, but also momentum flux, pressure, and shear stresses.

   Equation 4 – Energy-Momentum Tensor of an Ideal Fluid Formula: T^(mu_nu) = ((epsilon + p) / c^2) * u^mu * u^nu + p * g^(mu_nu) Condition: u^mu * u_mu = -c^2 Meaning of variables: epsilon (energy density in local rest frame), p (isotropic pressure), u^mu (four-velocity), g^(mu_nu) (inverse metric). Energy-oriented representation: The equation already uses energy density and

pressure. Further replacement of mass is not required for this form. Status: Established form for an ideal fluid. Limit: Not every matter or field configuration can be represented as an ideal fluid. Anisotropic stresses, electromagnetic fields, and other systems may require more specialized tensor forms. Relation to hypothesis: This equation is compatible with an energy-oriented description. However, it does not state that all forms of energy must microscopically be built from photons.

Equation 5 – Local Conservation of Energy and Momentum Formula: nabla_mu * T^(mu_nu) = 0 Meaning: The equation describes the local covariant conservation of energy and momentum of matter and field contents as far as they are contained in the considered total tensor T. Status: Established equation of GR. Restriction: Local covariant conservation does not mean that in an arbitrary dynamic spacetime, a globally defined total energy is always conserved or uniquely defined. Relation to hypothesis: An energy-oriented interpretation must respect this equation as long as established GR is to remain unchanged.

4. The Einstein Field Equations

   Equation 6 – Einstein Field Equations Formula: G_mu_nu + Lambda * g_mu_nu = (8 * pi * G / c^4) * T_mu_nu Without cosmological term: G_mu_nu = (8 * pi * G / c^4) * T_mu_nu Meaning: G_mu_nu is the Einstein tensor describing spacetime curvature. Status: Established fundamental equation of GR. Relation to hypothesis: The hypothesis adopts the field equations unchanged. The further interpretation that the geometric description could be a deeper-lying energy process is an additional ontological assumption.

   Equation 7 – Trace of the Einstein Field Equations Formula: R = 4 * Lambda - (8 * pi * G / c^4) * T (where T is the trace of the energy-momentum tensor) Meaning: Links scalar curvature R with trace T of the energy-momentum tensor. Status: Mathematical consequence of the field equations. Relation to hypothesis: It does not yet establish that curvature is physically caused by an additional energy flux.

   Equation 8 – Trace-Reduced Field Equations Formula: R_mu_nu = (8 * pi * G / c^4) * (T_mu_nu - 0.5 * T * g_mu_nu) + Lambda * g_mu_nu Status: Algebraically equivalent form of the Einstein field equations.

   Equation 9 – Vacuum Field Equations Formula: G_mu_nu + Lambda * g_mu_nu = 0 (respectively R_mu_nu = 0 for Lambda = 0) Meaning: Spacetime can be curved even without local matter and energy sources. Relation to hypothesis: An energy-oriented interpretation must explain how it handles vacuum solutions. From T_mu_nu = 0 it does not follow that every geometric structure disappears.

   Equation 10 – Bianchi Identity Formula: nabla_mu * G^(mu_nu) = 0 Status: Established geometric identity, consistent with nabla_mu * T^(mu_nu) = 0.

5. Interim Conclusion The first ten equations show three different cases:
6. Direct energy transformation: Mass and mass density can be expressed through energy equivalents.
7. Already energy-oriented equations: The energy-momentum tensor already contains the relevant quantities.
8. Geometric equations: Curvature tensors and vacuum equations cannot be "energy-oriented" simply by replacing mass. The central field equation remains unchanged. In this step, the hypothesis is an interpretation of the physical content, not a new theory. Continuation of the work series: Part II

covers the geodesic equation, motion of free bodies, and the transition to Newtonian gravitation, followed by the Schwarzschild solution, cosmology, and gravitational waves.

Part II: Equations of Motion, Conservation Laws, and Newtonian Limit

6. Motion in Spacetime The Einstein field equations describe the relationship between spacetime geometry and energy-momentum distribution. To calculate how bodies move, additional equations are required. In GR, freely falling test bodies unaffected by other forces move along geodesics, which describe the corresponding paths in spacetime.

   Equation 11 – Geodesic Equation Established form: d^2 x^mu / d tau^2 + Gamma^mu_{alpha beta} (dx^alpha / d tau) (dx^beta / d tau) = 0 Where:

- x^mu: the four spacetime coordinates,
- tau: the proper time of a massive test body,
- Gamma^mu_{alpha beta}: the Christoffel symbols calculated from the metric and its derivatives. Energy-oriented form: A direct replacement of mass with energy is not required. The mass of the freely falling test body does not appear explicitly in this equation. Status: Established equation of motion for freely falling test bodies under affine parametrization; for massive bodies, proper time is such a parametrization. Relation to hypothesis: The equation is particularly relevant because the motion of a test body does not depend directly on its rest mass. An energy-oriented interpretation could view the geometry as a description of motion conditions. However, the equation itself does not prove an additional energy-transfer mechanism.

  Equation 12 – Four-Velocity The four-velocity is defined as: u^mu = dx^mu / d tau For a massive body with signature (- + + +), it holds that: g_{mu nu} u^mu u^nu = - c^2 Meaning: The four-velocity describes the motion of a body in spacetime. Its norm is fixed for massive bodies under this convention. Energy-oriented form: No direct transformation needed. Status: Established definition and normalization condition. Relation to hypothesis: The equation uses a kinematic quantity and no explicit mass, making it independent of a specific representation of rest mass. Limit: For light and other massless particles, proper time along the path is not available as a suitable parameter; instead, an affine parameter is used, and the path is null (lightlike).

  Equation 13 – Four-Momentum For a massive body, the four-momentum is: p^mu = m u^mu Energy and spatial momentum are components of the four-momentum. In the local inertial frame, the familiar energy-momentum relation results: E^2 = p^2 c^2 + m^2 c^4 With rest energy E_0 = m c^2, it follows that: E^2 = p^2 c^2 + E_0^2 Meaning: Total energy depends on both rest energy and spatial momentum. Status: Established relativistic relationship; the second equation is a direct transformation. Relation to hypothesis: The energy form shows that mass can be replaced by rest energy in this relation without changing the physical content. Important restriction: This equation applies to the energy and momentum of a particle in a local inertial frame. It is not a general formula for a globally defined total energy in arbitrary curved spacetime.

  Equation 14 – Four-Acceleration The covariant four-acceleration is: a^mu = u^nu nabla_nu u^mu In coordinates: a^mu = du^mu / d tau + Gamma^mu_{alpha beta} u^alpha u^beta For a freely falling body on a geodesic: a^mu = 0 Meaning: Covariant four-acceleration describes deviation from free geodesic motion. Energy-oriented form: No mass replacement required. Status: Established definition and motion property of freely falling test bodies. Bezug zur Hypothese: In the GR model,

an ideal freely falling body feels no proper acceleration, even though its coordinate position may change. An alternative physical interpretation would have to reproduce this result.

7. How Neighboring Bodies Fall Differently Two bodies falling freely at different locations can accelerate relative to each other. In GR, this effect is described by spacetime curvature and observed as tidal action.

   Equation 15 – Geodesic Deviation Using a standard Riemann tensor convention, the equation can be written as: D^2 xi^mu / D tau^2 = - R^mu_{alpha nu beta} u^alpha xi^nu u^beta Where:

- xi^mu: the small separation vector between neighboring geodesics,
- R^mu_{alpha nu beta}: the Riemann curvature tensor,
- D / D tau: the covariant derivative along the reference path. Meaning: The relative acceleration of neighboring freely falling bodies depends on spacetime curvature. Energy-oriented form: There is no simple direct replacement of mass with energy; the equation describes a geometric effect. Status: Established equation of GR. Relation to hypothesis: This equation is central for investigating a possible energy-oriented interpretation, establishing a link between observable relative acceleration and the curvature tensor.

8. Conservation Quantities and Symmetries Not every spacetime possesses globally defined energy conservation. However, certain symmetries allow the construction of conserved quantities along geodesic paths.

   Equation 16 – Killing Vectors A vector field K^mu is called a Killing vector field if: nabla_mu K_nu + nabla_nu K_mu = 0 Meaning: Describes a continuous symmetry of the spacetime metric.

   Equation 17 – Conserved Quantity Along a Geodesic If K^mu is a Killing vector field and u^mu describes the four-velocity of a freely falling body, the quantity C = K_mu u^mu is constant along the geodesic. With four-momentum: C_p = K_mu p^mu Meaning: Temporal symmetry can lead to a conserved energy quantity; spatial rotational symmetry can yield a conserved angular momentum quantity.

9. Transition to Newtonian Gravitation Newtonian gravitation is an approximation of GR for weak gravitational fields and velocities small compared to the speed of light.

   Equation 18 – Newtonian Gravitational Potential of a Point Mass Established form: Phi(r) = - G M / r With energy equivalent E_0 = M c^2: Phi(r) = - G E_0 / (c^2 r) Meaning: Potential describes the weak gravitational field of a spherically symmetric source outside the source.

   Equation 19 – Gravitational Acceleration Newtonian equation of motion in the potential: a_vector = - nabla Phi For a spherically symmetric point source: a_vector = - (G M / r^2) r_unit_vector Energy-oriented form: a_vector = - (G E_0 / (c^2 r^2)) r_unit_vector

   Equation 20 – Poisson Equation of Newtonian Gravitation Continuous mass density: nabla^2 Phi = 4 pi G rho For non-relativistic matter (rho = epsilon / c^2): nabla^2 Phi = (4 pi G / c^2) epsilon

   Equation 21 – Newtonian Force Between Two Bodies Established form: F = G M_1 M_2 / r^2 Energy form: F = G E_{0,1} E_{0,2} / (c^4 r^2)

   Equation 22 – Newtonian Gravitational Potential Energy Established form: U(r) = - G M_1 M_2 / r Energy form: U(r) = - G E_{0,1} E_{0,2} / (c^4 r)

10. Weak Gravitational Field and Metric Equation 23 – Time Component of the Metric in a Weak Field g_{00} approx - (1 + 2 Phi / c^2) With energy potential: g_{00} approx - (1 - 2 G E_0 / (c^4 r))

    Equation 24 – Approximation of Motion in a Weak Field d^2 x_vector / dt^2 approx - nabla Phi = - (G E_0 / (c^2 r^2)) r_unit_vector

11. Escape Velocity and Energy Equation 25 – Newtonian Escape Velocity v_{esc} = sqrt(2 G M / r) Energy form: v_{esc} = sqrt(2 G E_0 / (c^2 r))

12. Summary of Part II

- Equations of motion do not always require explicit mass.
- The energy form is particularly easy to establish in the Newtonian limit.
- Not every geometric quantity thereby becomes an energy quantity.
- The central open question of the hypothesis remains: whether spacetime geometry physically emerges from a more fundamental energy process.

    Part III: Schwarzschild Solution, Event Horizon, and Black Holes

13. Classification of Schwarzschild Solution The Schwarzschild solution is an exact solution of the Einstein field equations describing spacetime outside a spherically symmetric, non-rotating, uncharged body. Relevant for the hypothesis is that the Schwarzschild mass parameter M can be expressed via its energy equivalent E_0 = M * c^2.

14. Schwarzschild Radius Equation 26 (Established): r_s = (2 * G * M) / c^2 Equation 27 (Energy form): r_s = (2 * G * E_0) / c^4

15. Schwarzschild Metric Equation 28 (Established): ds^2 = -(1 - (2 * G * M) / (c^2 * r)) * c^2 * dt^2 + (1 - (2 * G * M) / (c^2 * r))^(-1) * dr^2 + r^2 * dOmega^2 Equation 29 (Energy form): ds^2 = -(1 - (2 * G * E_0) / (c^4 * r)) * c^2 * dt^2 + (1 - (2 * G * E_0) / (c^4 * r))^(-1) * dr^2 + r^2 * dOmega^2

16. Metric Components and Spatial Dependence Equation 30: g_tt = -(1 - (2 * G * E_0) / (c^4 * r)) Equation 31: partial g_tt / partial r = -(2 * G * E_0) / (c^4 * r^2)

17. Event Horizon Equation 32: r = r_s = (2 * G * E_0) / c^4

18. Gravitational Time Dilation Equation 33: dtau = sqrt(1 - (2 * G * E_0) / (c^4 * r)) * dt

19. Light and Gravitational Redshift Equation 34: Frequency ratio expressed with E_0 parameters.

20. Spacetime Curvature and Physical Singularity Equation 35 (Kretschmann scalar): K = (48 * G^2 * E_0^2) / (c^8 * r^6)

21. Horizon Area Equation 36: A = (16 * pi * G^2 * E_0^2) / c^8

22. Hawking Temperature Equation 37: T_H = (hbar * c^5) / (8 * pi * G * E_0 * k_B)

23. Summary of Part III Central relationships of the Schwarzschild solution can be systematically expressed in energy form using E_0.

    Part IV: Rotating and Charged Black Holes and Further Solutions

24. Rotating Black Holes: Kerr Solution Equation 38: E_0 = M * c^2 <=> M = E_0 / c^2 Equation 39 (Rotation parameter): a = (J * c) / E_0 Equation 40 (Radial function): Delta = r^2 - (2 * G * E_0 *

r) / c^4 + (J * c / E_0)^2 Equation 41 (Outer horizon radius): r_+ = (G * E_0) / c^4 + sqrt(((G * E_0) / c^4)^2 - (J * c / E_0)^2)

25. Electromagnetic Energy Equation 42 (Electric field energy density): u_el = 0.5 * epsilon_0 * |E|^2 Equation 43 (Electric field energy): U_el = Integral over V of (0.5 * epsilon_0 * |E|^2) dV Equation 44 (Outer field energy of a point charge): U_outer(R) = Q^2 / (8 * pi * epsilon_0 * R) Equation 45 (Total energy-momentum tensor): T_mu_nu^{total} = T_mu_nu^{matter} + T_mu_nu^{EM}

26. Charged Black Holes: Reissner-Nordström Solution Equation 46 (Radial function): f(r) = 1 - (2 * G * E_0) / (c^4 * r) + (G * Q^2) / (4 * pi * epsilon_0 * c^4 * r^2) Equation 47 (Horizon radii): r_± = (G * E_0) / c^4 ± sqrt(((G * E_0) / c^4)^2 - (G * Q^2) / (4 * pi * epsilon_0 * c^4)) Equation 48 (Condensed charge term): r_Q^2 = (G * Q^2) / (4 * pi * epsilon_0 * c^4)

27. Rotating and Charged Black Holes: Kerr-Newman Solution Equation 49 (Radial function): Delta = r^2 - (2 * G * E_0 * r) / c^4 + (J * c / E_0)^2 + (G * Q^2) / (4 * pi * epsilon_0 * c^4) Equation 50 (Outer event horizon): r_+ = (G * E_0) / c^4 + sqrt(((G * E_0) / c^4)^2 - (J * c / E_0)^2 - (G * Q^2) / (4 * pi * epsilon_0 * c^4))

28. Conservation of Electric Charge Equation 51 (Dimensions): [E_0] = J and [Q] = C Equation 52 (Local charge conservation): nabla_mu * j^mu = 0

29. Summary of Part IV Equations for rotating and charged black holes are successfully represented in energy-equivalent terms for mass parameters, while angular momentum, electric charge, and electromagnetic field quantities retain their independent roles.